Econometrics

MFE5340 - AI in FE: Quantitative Investment

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1 Introduction

Academia considers tradable portfolios as factors. Therefore, for any selected firm characteristic, we need to construct a portfolio based on it. This portfolio is referred to as the *factor-mimicking portfolio* and the return of this portfolio represents the return of the factor. The most common method for constructing portfolios is the **portfolio sort** technique introduced by Fama and French.

Let's take the classic valuation indicator, the book-to-market ratio (BM), also known as the inverse of the price-to-book ratio (P/B), as an example to explain the method. Since BM is a valuation indicator, the factor constructed from it is referred to as the value factor. In portfolio sort, the variable used to rank stocks (such as BM) is called the *sorting variable*. **The core idea of the method is to use the stock's BM as a proxy for its exposure to the factor.** It is important to note that this method does not assume that the variable value equals factor exposure, nor does it assume any specific mathematical relationship between the two. The method simply assumes that the variable is correlated with the factor exposure. For example, with BM, the method assumes that stocks with high BM have higher exposure to the value factor, and those with low BM have lower exposure, and that's it. Under this idea, even though we do not know the exact factor exposure of individual stocks, we can use the variable ranking to approximate it and construct the factor-mimicking portfolio.

Let's now discuss the procedure (See Figure 1). Portfolio sort involves three steps:

1. **Sorting:** First, define the stock pool and rank all the stocks in the pool based on the sorting variable (BM in this example) from high to low, or low to high, depending on the relationship between the sorting variable and stock return (i.e., positively or negatively correlated). (If the variable is negatively correlated with returns, the stocks are sorted from low to high. The low-volatility factor is an example of this.)

2. **Grouping:** Divide all stocks into L groups based on their ranking (by convention, the stocks are divided into 10 groups based on deciles of the sorting variable, so $L = 10$). A long/short hedged portfolio is then constructed by going long the stocks in the highest-ranked group and shorting the stocks in the lowest-ranked group. This portfolio is referred to as a spread portfolio. By construction, the returns of the long and short legs of the spread portfolio correspond to the returns of the top $1/L$ of stocks with the highest ranks and the bottom $1/L$ of stocks with the lowest ranks. The difference between these returns reflects the return of the factor constructed around that variable. Therefore, the spread portfolio represents the factor-mimicking portfolio, and the spread portfolio's return is the factor's return. It is important to note that when constructing the spread portfolio, the long and short legs are typically dollar neutral, meaning that both the long and short legs have equal capital. Additionally, since both portfolios contain multiple stocks, it is necessary to choose a weighting method for the stocks, with the most common approaches being value weighting (i.e., weighting by market

capitalization) and equal weighting.

3. **Rebalance:** Since the value of the sorting variable for individual stocks changes over time (**indicating that their exposure to the factor also changes over time** and this is important), it is necessary to periodically repeat the above two steps to rebalance the factor-mimicking portfolio. The rebalancing frequency is typically monthly or even annually in academic studies. (But in industry practice, it is not uncommon to rebalance a factor portfolio daily.) Repeating this process over time generates the time series of factor returns.

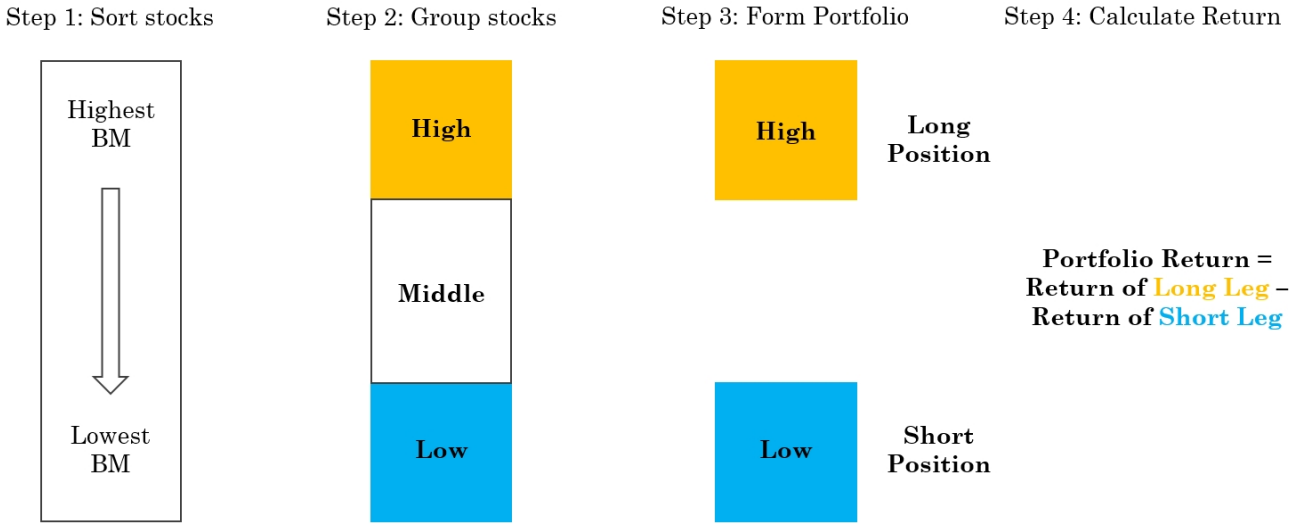


Figure 1: Portfolio Sort Procedure Illustration

2 Portfolio Sort Test

Portfolio sort allows people to easily construct factor-mimicking portfolios and obtain the returns of the L portfolios as well as the factor portfolio. Once these data are available, two primary tests can be conducted: testing the factor's risk premium and testing the monotonicity of the L portfolios' returns. In academic literature, these tests are referred to as portfolio sort tests.

2.1 Testing Factor's Risk Premium

The primary objective of the portfolio sort test is to examine the expected excess return (or risk premium) of the factor. In factor research, the null hypothesis is typically that the factor's expected excess return is zero. The test focuses on whether the average return calculated from sample data can reject the null hypothesis at a given significance level.

Let $\{\lambda_t\}, t = 1, 2, \cdots, T$ represent the time series of factor returns, then the estimated risk

premium $\hat{\lambda}$ and its standard error s.e.($\hat{\lambda}$) are given by:

$$\hat{\lambda} = \frac{1}{T}\sum_{t=1}^{T}\lambda_t, \tag{1}$$

$$\text{s.e.}(\hat{\lambda}) = \frac{\text{std}(\lambda_t)}{\sqrt{T}}, \tag{2}$$

where std(λ_t) is the sample standard deviation of λ_t. With $\hat{\lambda}$ and s.e.($\hat{\lambda}$), we can construct the t-statistic under the null hypothesis (i.e., $\lambda = 0$):

$$t\text{-statistic} = \frac{\hat{\lambda}}{\text{s.e.}(\hat{\lambda})}. \tag{3}$$

This follows a t-distribution with $T-1$ degrees of freedom (why $T-1$? because we adjust the degree of freedom when calculating the sample standard deviation!). Using the t-statistic, a (two-tailed) p-value can be calculated, and based on the t-statistic or p-value, we can determine whether to accept or reject the null hypothesis at the given significance level.

By convention, in academia, significance levels of 0.05 and 0.01 are commonly used, corresponding to t-statistic thresholds of approximately 2.0 and 2.6 for large samples, respectively. Thus, if the t-statistic calculated from the sample data exceeds 2.0, the result is considered significant under the null hypothesis. If the null hypothesis cannot be rejected, then the factor under study does not satisfy the requirement of having an expected excess return greater than zero.

2.2 Monotonicity

In addition, since a good factor should be able to explain the cross-sectional variation in individual stock excess returns, the second issue that the method addresses is whether the returns of the L portfolios formed based on the sorting variable exhibit strong monotonicity. This can be tested by calculating the **rank correlation coefficient** between the portfolio returns and the sorting variable groups. The rank correlation coefficient, similar to the regular correlation coefficient, is computed by converting the original values into ranks, and it measures the *monotonic relationship* between two random variables. In this regard, we can use the Spearman rank correlation coefficient.

Let X_r denote the ranks of portfolios' returns, and X_g note the sorting variable groups. Then, the rank correlation coefficient ρ_s is:

$$\rho_s = \frac{\text{cov}(X_r, X_g)}{\sigma_{X_r}\sigma_{X_g}}. \tag{4}$$

It is evident that when the returns of the L portfolios increase monotonically with the sorting variable groups, the rank correlation coefficient will be 1.0. Conversely, if the returns decrease monotonically with the sorting variable groups, the rank correlation coefficient is -1.0. The range

of ρ_s is therefore from -1.0 to 1.0.

2.3 An Example

Let's continue with the BM (book-to-market ratio) example to illustrate the tests discussed above. Using portfolio sort, **at each month t (i.e. each cross-section)**, stocks in the Chinese stock market are divided into $L = 10$ groups (denoted as Low, 2, $\cdots$, 9, and High, where Low represents the group with the lowest BM values, and High represents the group with the highest BM values).

Stocks in each group are weighted by market capitalization (i.e., value weighting), and the portfolio is rebalanced monthly. In addition, a factor portfolio is constructed by going long the highest BM group (the High group) and shorting the lowest BM group (the Low group). The following table presents the average returns of different groups as well as the factor (i.e., the High - Low group). From the results, we can see that the factor's monthly average return is 0.88%, with a standard error of 0.47%, a t-statistic of 1.85, and a p-value of 0.07, indicating that we can reject the null hypothesis at the 0.1 significance level.

Table 1: Portfolio Sort Results using BM

	Low	1	2	3	4	5	6	7	8	High	High − Low
Average Return (%)	0.57	0.53	0.57	0.72	0.94	0.90	1.08	1.32	1.21	1.44	0.88
Std Error (%)	0.57	0.56	0.55	0.58	0.56	0.53	0.57	0.58	0.56	0.58	0.47
t-statistic	1.00	0.95	1.03	1.24	1.69	1.69	1.90	2.27	2.15	2.48	1.85
p-value	0.32	0.35	0.30	0.22	0.09	0.09	0.06	0.02	0.03	0.01	0.07

Figure 2 shows the monthly average returns of the 10 portfolios. Their average returns generally increase with BM. The rank correlation coefficient is 0.94 (with a p-value of 0.00), indicating a significant monotonic relationship between the two.

Due to its simplicity and ease of use, portfolio sort has been widely applied in both academic and industry research on style factors. It is worth mentioning that portfolio sort mentioned earlier is based on a single variable, and therefore it is also referred to as *univariate sorting*.

It is important to remember that univariate sorting is a relatively crude way to construct factors. Ideally, a factor-mimicking portfolio should have no exposure to other factors, but univariate sorting struggles to control the influence of other factors. For example, if all high-BM stocks are large-cap stocks and all low-BM stocks are small-cap stocks, then the factor portfolio will simultaneously go long large-cap stocks and short small-cap stocks. As a result, although the spread portfolio is constructed by using BM, its returns would be influenced by both BM and market cap. To minimize interference from other factors, it is common to use more than one variable for double sorting or triple sorting, collectively known as multivariate sorting.

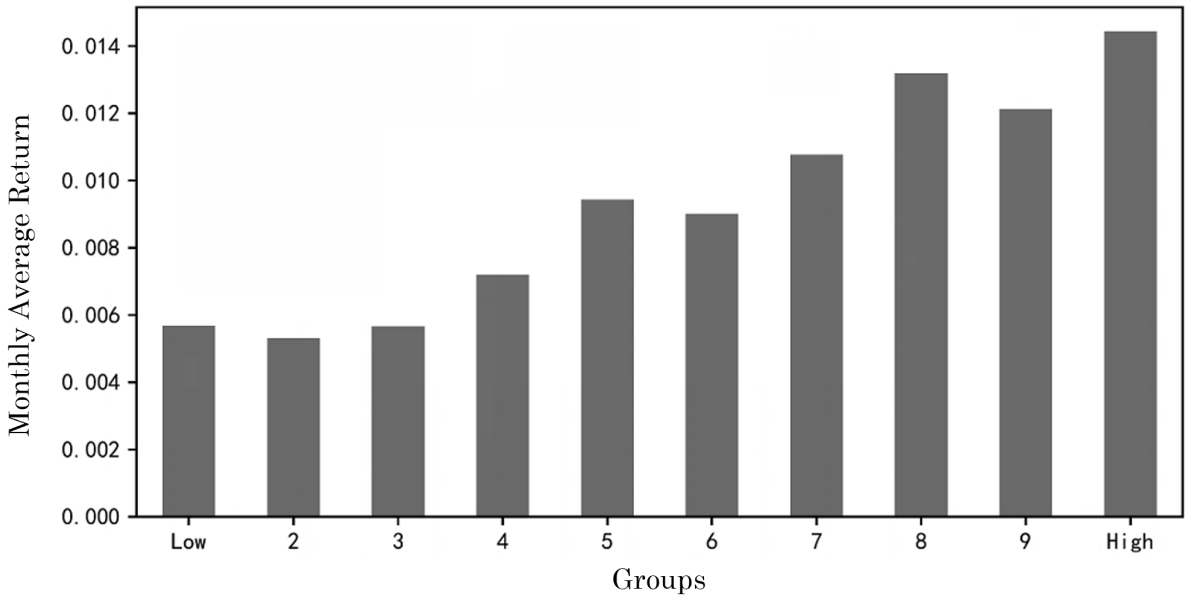


Figure 2: Portfolio Sort using BM

3 Double Sorting

The most common multivariate sorting is double sorting (also known as bivariate sorting). Double sorting involves sorting and constructing factor-mimicking portfolios based on two variables. Consider variables X_1 and X_2, where the stocks are divided into L_1 and L_2 groups based on these two variables, resulting in a total of $L_1 \times L_2$ combinations. One key detail in double sorting is whether the two variables are sorted independently or if one variable depends on the other. The former is called *independent* double sorting (or *unconditional* double sorting), while the latter is referred to as *dependent* double sorting (or *conditional* double sorting).

3.1 Independent Double Sorting

Let's first consider the case of independent double sorting. A common choise for L_1 and L_2 is to use quintiles for each variable. Suppose two sorting variables are used to independently divide the stocks into five groups as shown in the following figure, forming a total of 25 portfolios. The weights of stocks in these 25 portfolios can be equal-weighted or value-weighted. From these 25 portfolios, factor-mimicking portfolios can be constructed for a given variable.

Now, take X_1 as an example. The idea is similar to univariate sorting: go long on stocks ranked high on the variable and short stocks ranked low. Since double sorting is applied, for each group of X_2, there is a High X_1 portfolio. In other words, there are $L_2(= 5)$ such portfolios, i.e., $P_{51}, P_{52}, P_{53}, P_{54}, P_{55}$. Similarly, there are $L_2(= 5)$ Low X_1 portfolios, and they are $P_{11}, P_{12}, P_{13}, P_{14}, P_{15}$. Finally, we go long the five High portfolios (with equal weighting) and go short the five Low portfolios (with equal weighting). This leads to the factor portfolio for X_1. Let

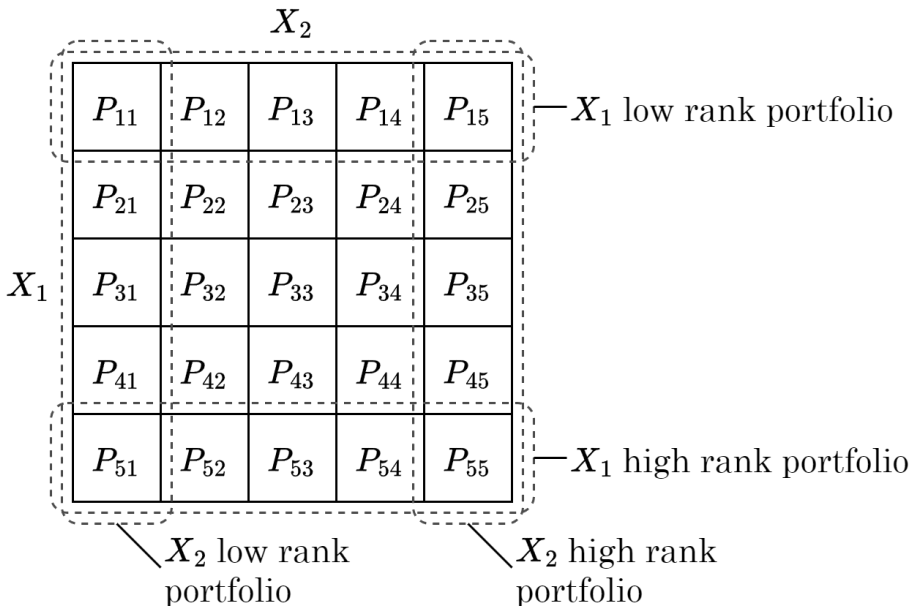


Figure 3: Portfolio Sort using BM

$R_{ij,t}$ represent the return of portfolio P_{ij} in period t, then the return of the factor $\lambda_{X_1 t}$ is:

$$\lambda_{X_1 t} = \frac{1}{L_2} \sum_{i=1}^{L_2} R_{L_1 i,t} - \frac{1}{L_2} \sum_{i=1}^{L_2} R_{1i,t}. \tag{5}$$

In independent double sorting, the two variables are treated symmetrically, making it straightforward to construct a factor portfolio based on X_2. This portfolio is formed by going long on the $L_1(= 5)$ High X_2 portfolios, i.e., $P_{15}, P_{25}, P_{35}, P_{45}, P_{55}$, and going short the $L_1(= 5)$ Low X_2 portfolios, i.e., $P_{11}, P_{21}, P_{31}, P_{41}, P_{51}$. The return of this factor, denoted by $\lambda_{X_2 t}$, is:

$$\lambda_{X_2 t} = \frac{1}{L_1} \sum_{i=1}^{L_1} R_{iL_2,t} - \frac{1}{L_1} \sum_{i=1}^{L_1} R_{i1,t}. \tag{6}$$

Once we have the time series of factor returns $\{\lambda_{X_1 t}\}$ and $\{\lambda_{X_2 t}\}$, we can derive the t-statistics for them and test their risk premiums. It's worth mentioning that by rearranging the portfolio returns in above equations, we can obtain an alternative, equivalent form:

$$\lambda_{X_1 t} = \frac{1}{L_2} \sum_{i=1}^{L_2} \left(R_{L_1 i,t} - R_{1i,t} \right), \tag{7}$$

and

$$\lambda_{X_2 t} = \frac{1}{L_1} \sum_{i=1}^{L_1} \left(R_{iL_2,t} - R_{i1,t} \right). \tag{8}$$

Taking the first one as an example, it means that for each group i of the X_2 variable, we select the X_1 High and Low portfolios and calcuate their return differnece $R_{L_1 i} - R_{1i}$. Then, we average the differences across all L_2 (which is 5 in this example) groups to obtain $\lambda_{X_1 t}$. Apparently, these

two approaches are equivalent.

In academic research on anomalies, double sorting is often used. To test whether a new anomaly variable can generate excess returns, scholars typically perform double sorting using the anomaly variable alongside an established factor variable to eliminate the effects of existing factors. During the analysis, in addition to focusing on the anomaly's returns, researchers also examine whether the anomaly variable can distinguish cross-sectional differences in stock returns within each group of the established factor.

While independent double sorting is simple, it has a drawback: independent sorting can lead to certain portfolios containing too few stocks. For example, suppose there are 1,000 stocks, and we sort them into 5 groups based on both X_1 and X_2, resulting in 25 combinations, with an average of 40 stocks per group. If the *cross-sectional* correlation between X_1 and X_2 is high, a stock with a high X_1 value is also likely to have a high X_2 value. This would cause the groups where both variables are similar—those along the diagonal the above figure —to have more stocks, while other groups (particularly high X_1 but low X_2, and low X_1 but high X_2 groups) would contain fewer stocks. This imbalance in the number of stocks within groups can make the factor returns more susceptible to outliers. While this drawback exists in factor research, it typically does not cause significant issues, as variables with high cross-sectional correlation are usually not chosen for double sorting. If the correlation between the two sorting variables is unclear, we can also count the number of stocks in each group. If all $L_1 \times L_2$ portfolios contain a sufficient number of stocks, there is generally no major concern.

3.2 Dependent Double Sorting

Next, we introduce dependent or conditional double sorting. In dependent sorting, the two variables are applied in a **specific order** for sorting the stocks. Suppose we first sort the stocks based on X_1, dividing all stocks into L_1 groups. Then, *within* each of these L_1 groups, we further sort the stocks based on X_2, dividing the stocks into L_2 groups within each X_1 group. This again results in a total of $L_1 \times L_2$ groups. From this example, it becomes clear that dependent double sorting examines the effect of X_2 on stock returns after controlling for X_1. Conversely, if the sorting order is X_2 first and then X_1, it analyzes the impact of X_1 on stock returns after controlling for X_2.

Dependent double sorting focuses on whether the second variable provides *incremental* information in explaining returns after the first variable is controlled for. In this method, the two sorting variables are not treated equally: the first sorting variable acts solely as a control variable, and the main interest is the relationship between the second sorting variable and asset returns. Therefore, a factor should only be constructed based on the second sorting variable, and its factor returns should be calculated accordingly.

Academia typically employs two approaches to calculate factor returns for the second sorting variable. The first approach is identical to the return formula used in independent double sorting. Let's focus on the second approach. Assuming X_1 and X_2 are the first and second sorting variables,

respectively, then we take the *union* of all the High X_2 groups, $P_{iL_2}, i = 1, 2, \cdots, L_1$, and the *union* of all the Low X_2 groups, $P_{i1}, i = 1, 2, \cdots, L_1$:

$$P_{L_2}^{\text{top}} = P_{1L_2} \cup P_{2L_2} \cup \cdots \cup P_{L_1L_2}, \tag{9}$$

$$P_{L_2}^{\text{bottom}} = P_{11} \cup P_{21} \cup \cdots \cup P_{L_11}. \tag{10}$$

All the stocks in $P_{L_2}^{\text{top}}$ are weighted, either by market value or equally, to form the long leg for X_2. Similarly, the stocks in $P_{L_2}^{\text{bottom}}$ are weighted in the same way to form the short leg. In each period t, the return on the long leg (denoted as $R_{L_2}^{\text{top}}$) minus the return on the short leg (denoted as $R_{L_2}^{\text{bottom}}$) gives the factor return:

$$\lambda_{X_2 t} = R_{L_2}^{\text{top}} - R_{L_2}^{\text{bottom}}. \tag{11}$$

The two methods for calculating the factor return of X_2 (i.e., the second sorting variable) are slightly different, but upon closer examination, it becomes clear that when stocks in each portfolio are equally weighted, they are equivalent. The difference only arises when stocks in each portfolio are weighted unequally, such as by market cap.

Finally, it is worth noting that dependent double sorting ensures that each group contains a sufficient number of stocks. In the example where $L_1 = L_2 = 5$, if these five groups are divided according to the quintiles of the sorting variables, the 25 resulting portfolios will contain an equal number of stocks. In more general cases, the number of groups divided by the two variables may differ, and the quantiles used to divide the groups may vary, resulting in different numbers of stocks in each group, but it still guarantees that there will be enough stocks in each group.

3.3 Two Comments

There are two additional points to clarify regarding double sorting. The purpose of double sorting is to eliminate the mutual influence between two variables, allowing for a more accurate calculation of the returns of the factor-mimicking portfolios constructed around each variable. In this regard, dependent double sorting is a better choice than independent double sorting, as it examines the *conditional* relationship between the second sorting variable and stock returns after controlling for the first variable.

However, in academic research, independent double sorting is more commonly used when studying factors, possibly because the classic Fama and French (1993) three-factor model initially employed independent double sorting. On the other hand, when studying anomalies, it is not uncommon to use dependent double sorting with market cap as the conditional variable (i.e., the first sorting variable) and the anomaly variable to construct anomaly portfolios and calculate their excess returns. In such studies, both methods for calculating returns are used (see e.g., Bali et al. 2014 and Liu et al. 2019).

The second clarification concerns the number of groups used when performing double sorting. In anomaly research, when the stock pool is large, a 5×5 or 10×10 sorting is common. However, in factor research, a 2×3 division is often used, with market cap being one of the variables. This practice is also linked to the Fama and French (1993) three-factor model. Since this paper is the foundation of empirical work, many of its methods have had a profound influence on academia. In constructing the size (SMB) and value (HML) factors, the authors employed independent double sorting, using market cap and BM (book-to-market ratio). Stocks were divided into two groups based on size (Small and Large) and three groups based on BM (Low, Medium and High), forming six portfolios from the $2 \times 3 = 6$ combinations. This approach, i.e. the 2×3 independent double sorting with market cap and another variable has since been widely adopted following Fama and French (1993).

4 Fama-MacBeth Regression

In 1973, Eugene Fama and James MacBeth introduced a two-step regression method in their paper Fama and MacBeth (1973), known as the Fama-MacBeth regression. The goal of their paper was to test the CAPM, but the method gained wide acceptance in the industry because it cleverly accounts for the effect of cross-sectional correlation of residuals on standard errors. This paper is one of the most cited papers in econometrics.

Similar to cross-sectional regression, the first step in the Fama-MacBeth regression is also to estimate the factor exposures $\hat{\boldsymbol{\beta}}_i$ for each asset i from N time-series regressions. The key difference between Fama-MacBeth regression and cross-sectional regression lies in the second step. In cross-sectional regression, the time-series average returns $\mathbb{E}_T[R_i^e]$ are regressed on $\hat{\boldsymbol{\beta}}_i$ in a **single** cross-section. However, in the 2nd step of Fama-MacBeth regression, we regress the excess return R_{it}^e for period t on the factor exposure estimated at $t-1$, resulting in T cross-sectional regressions. This is the primary difference between Fama-MacBeth and cross-sectional regression tests. For time t, the cross-sectional regression model is:

$$R_{it}^e = \hat{\boldsymbol{\beta}}_i^\top \boldsymbol{\lambda}_t + \alpha_{it}, \quad i = 1, 2, \ldots, N. \tag{12}$$

If an intercept term is included, the model becomes:

$$R_{it}^e = \gamma_t + \hat{\boldsymbol{\beta}}_i^\top \boldsymbol{\lambda}_t + \alpha_{it}, \quad i = 1, 2, \ldots, N. \tag{13}$$

Comparing cross-sectional regression and Fama-MacBeth regression, in the former, the time-series average $\mathbb{E}_T[R_{it}^e]$ is calculated first, and it is then used in a single cross-sectional regression with $\hat{\boldsymbol{\beta}}_i$ being independent variable. In contrast, Fama-MacBeth regression estimates the regression model at each t (e.g., with $T = 100$ periods of data, 100 cross-sectional regressions are performed), obtaining estimates of factor returns and residuals (pricing errors) $\hat{\boldsymbol{\lambda}}_t$ and $\hat{\alpha}_{it}$. **Fama-MacBeth**

then averages the results of the T total cross-sectional regressions to obtain the estimates for the factor risk premiums and the pricing errors for each asset i:

$$\hat{\boldsymbol{\lambda}} = \frac{1}{T}\sum_{t=1}^{T}\hat{\boldsymbol{\lambda}}_t, \tag{14}$$

$$\hat{\alpha}_i = \frac{1}{T}\sum_{t=1}^{T}\hat{\alpha}_{it}. \tag{15}$$

The ingenuity of the Fama-MacBeth regression lies in treating the regression results from T periods as T independent samples. Traditional cross-sectional regression tests only perform one regression, yielding a single sample estimate of factor returns and pricing errors. In contrast, the Fama-MacBeth approach provides time series of factor returns $\{\hat{\lambda}_{kt}\}$ and asset pricing errors $\{\hat{\alpha}_{it}\}$, making it straightforward to compute the standard errors:

$$\sigma(\hat{\lambda}_k) = \left[\frac{1}{T^2}\sum_{t=1}^{T}\left(\hat{\lambda}_{kt} - \hat{\lambda}_k\right)^2\right]^{1/2}, \tag{16}$$

$$\sigma(\hat{\alpha}_i) = \left[\frac{1}{T^2}\sum_{t=1}^{T}\left(\hat{\alpha}_{it} - \hat{\alpha}_i\right)^2\right]^{1/2}. \tag{17}$$

Note that the coefficient above is $\frac{1}{T^2}$ rather than $\frac{1}{T}$ because we are finding standard errors of sample means.

In matrix form, the covariance matrix of all N pricing errors is:

$$\hat{\boldsymbol{\alpha}} = \frac{1}{T}\sum_{t=1}^{T}\hat{\boldsymbol{\alpha}}_t, \tag{18}$$

$$\operatorname{cov}(\hat{\boldsymbol{\alpha}}) = \frac{1}{T^2}\sum_{t=1}^{T}\left(\hat{\boldsymbol{\alpha}}_t - \hat{\boldsymbol{\alpha}}\right)\left(\hat{\boldsymbol{\alpha}}_t - \hat{\boldsymbol{\alpha}}\right)^\top. \tag{19}$$

Therefore, a χ^2-statistic can be used to test these pricing errors jointly:

$$\hat{\boldsymbol{\alpha}}^\top \operatorname{cov}(\hat{\boldsymbol{\alpha}})^{-1}\hat{\boldsymbol{\alpha}} \sim \chi^2_{N-K}. \tag{20}$$

Similarly, with the estimates and standard errors of $\hat{\lambda}_k$ and $\sigma(\hat{\lambda}_k)$, we can construct the t-statistics to test the factor risk premiums.

Let's recap. The key difference between the 2nd step (i.e., cross-sectional regression step) in Fama-MacBeth and traditional cross-sectional regression tests is as follows:

- In Fama-MacBeth regression, OLS (or GLS) is first used at each time t to estimate factor returns and pricing errors at t (by regressing R^e_{it} on the estimated factor exposures), and then the time-series averages of the estimates $\hat{\boldsymbol{\lambda}}_t$ and $\hat{\alpha}_{it}$ are computed to obtain $\hat{\boldsymbol{\lambda}} = \mathbb{E}_T[\hat{\boldsymbol{\lambda}}_t]$ and $\hat{\alpha}_i = \mathbb{E}_T[\hat{\alpha}_{it}]$.

- In traditional cross-sectional regression, the time-series averages of R_{it}^e, i.e., $\mathbb{E}_T[R_{it}^e]$, are taken first, and then OLS is used for estimation, directly yielding $\hat{\boldsymbol{\lambda}}$ and $\hat{\alpha}_i$.

In simple terms, Fama-MacBeth regression "estimates first, then averages", whereas traditional cross-sectional regression "averages first, then estimates". Thus, Fama-MacBeth regression can be considered a special form of cross-sectional regression. **The advantage of Fama-MacBeth regression is that it can eliminate (or reduce) the impact of cross-sectional correlation of errors on the standard error.** If factor exposures remain the same across all time periods, then these two methods are identical. (Why?) If the factor exposures $\hat{\boldsymbol{\beta}}_i$ do not change over time (i.e., they are constant across all periods), then each cross-sectional regression in the Fama-MacBeth procedure would essentially use the same exposures as in the single cross-sectional regression of the traditional method. Therefore, averaging the results of T cross-sectional regressions in Fama-MacBeth would yield the same results as a single cross-sectional regression using the average returns and factor exposures in the traditional method.

In practical applications, due to the flexibility of the Fama-MacBeth method, it usually assumes that the factor exposures $\boldsymbol{\beta}_i$ do not remain constant across all T periods. In fact, in the original Fama and MacBeth (1973) paper, the authors used a rolling window when estimating $\boldsymbol{\beta}_i$ in the first step (i.e., the time-series regression step), allowing the factor exposure vector to **vary** over time t. Specifically, for period t, the factor exposures $\boldsymbol{\beta}_{it-1}$ are estimated using historical data from a given window up to period $t-1$. Since the estimate is based on data up to $t-1$, the time subscript of the factor exposure vector is $t-1$. The estimated $\hat{\boldsymbol{\beta}}_{it-1}$ is then used as the explanatory variable in the cross-sectional regression for period t, leading to the following cross-sectional regression model:

$$R_{it}^e = \hat{\boldsymbol{\beta}}^\top{}_{it-1}\boldsymbol{\lambda}_t + \alpha_{it}, \quad i = 1, 2, \cdots, N, \tag{21}$$

or alternatively, with an intercept:

$$R_{it}^e = \gamma_t + \hat{\boldsymbol{\beta}}^\top{}_{it-1}\boldsymbol{\lambda}_t + \alpha_{it}, \quad i = 1, 2, \cdots, N. \tag{22}$$

Now, let's discuss some limitations of the Fama-MacBeth regression. First, it cannot handle the time-series correlation in α_{it}. However, it is generally believed that the time-series correlation of stock returns is weak, so Fama-MacBeth regression does not encounter significant issues. Second, since the factor exposures $\hat{\boldsymbol{\beta}}_i$ used in the cross-sectional regression are estimated from time-series regressions, they also contain errors (error-in-variables issue). Therefore, Shanken correction is still required. Despite these limitations, Fama-MacBeth regression has been widely recognized in academia. To this day, approximately one-third of econometric papers analyzing panel data still use Fama-MacBeth regression, and it remains a standard technique in almost every asset pricing study. It is also widely adopted in industry practice where firm characteristics are used as factor exposures. (You will be asked to perform Fama-MacBeth regression in your homework.)

5 Testing Anomalies

The study of anomalies has been a subject of intense academic interest. Over the past thirty years, scholars have discovered hundreds of anomalies in the U.S. stock market. The paper by Hou et al. (2020) invested significant effort in replicating over 450 anomalies, providing a comprehensive summary. Unfortunately, they found that the majority of these published anomalies could not generate significant α against factor models. This implies that much of the academic research has not yielded scientifically sound and correct results.

Before diving into specific tests, it's worth mentioning an important trend in anomaly research. In the past, the most common approach was to use a single variable to construct anomalies. However, in recent years, a mainstream trend has emerged: using multiple (i.e., composite) variables to construct anomalies. For example, in the paper by Asness et al. (2019), titled *Quality Minus Junk*, the authors introduced a quality anomaly. They used numerous variables across different categories to derive an overall score for stock ranking and constructed an anomaly portfolio. Another example is the F-score of Piotroski (2000) that you have seen in your assignment.

Of course, this approach has its pros and cons. On the positive side, these anomalies constructed with multiple variables often carry stronger financial implications, which can help people better understand the relationship between firm characteristics and assets expected returns. On the downside, using composite variables can increase the *risk of overfitting*. In a sense, searching for anomalies (and factors) is a form of data mining by using historical data. When too many variables are used, there's a greater chance of over-mining, leading to the so-called anomalies being merely the result of in-sample overfitting.

Now, let's explore how anomalies are tested.

5.1 Time-Series Regression

As you may recall, if an asset is able to generate significant excess returns (i.e., α) that cannot be explained by a factor model, it is considered an anomaly. Anomalies are typically constructed as portfolios by following these steps:

1. Select a potential financial or technical variable as the anomaly variable.
2. Use portfolio sort to construct the anomaly portfolio using that variable and derive the time series of anomaly returns.
3. Test whether the anomaly generates significant α against a given factor model.

The first approach for testing anomalies, as you may imagine, is time-series regression. Let $R_t^e, t = 1, 2, \cdots, T$ represent the anomaly returns at time t, and let $\boldsymbol{\lambda}_t, t = 1, 2, \cdots, T$ represent the vector of factor returns at time t (assuming there are K factors). When testing for an anomaly,

the null hypothesis is that there is no unexplained portion of anomaly returns by the factors, i.e., $\alpha = 0$. To test this, we regress R_t^e on $\boldsymbol{\lambda}_t$:

$$R_t^e = \alpha + \boldsymbol{\beta}_a^\top \boldsymbol{\lambda}_t + \varepsilon_t, \quad t = 1, 2, \cdots, T. \tag{23}$$

This model can be estimated via OLS. The estimated intercept, $\hat{\alpha}$, represents the portion of anomaly returns that cannot be explained by the factor model. Besides, by checking the estimated factor exposures $\hat{\boldsymbol{\beta}}_a$, we will be able to tell which factors contribute to explaining the anomaly return. Since the goal is to test whether the anomaly produces significant excess returns, we use $\hat{\alpha}$ and its standard error s.e.($\hat{\alpha}$) to construct the t-statistic (d.o.f $= T - K - 1$):

$$t\text{-statistic} = \frac{\hat{\alpha}}{\text{s.e.}(\hat{\alpha})}. \tag{24}$$

In the time-series regression, the total number of periods is T. On the other hand, the independent variables include an intercept and K factors, resulting in a total of $K + 1$ independent variables. Therefore, the degrees of freedom for the t-statistic is $T - K - 1$. Based on the t-statistic, we can compute the p-value to decide whether to reject the null hypothesis at a given significance level. If the null hypothesis $\alpha = 0$ is rejected, then the LHS portfolio is considered an anomaly.

There is an econometric issue that must be addressed in time-series regressions. When the random errors in the time series exhibit autocorrelation or heteroscedasticity, the standard errors from OLS become inaccurate, leading to distorted t-statistics. We discuss this issue next.

5.2 Econometric Issues

Consider the following linear regression model for the population:

$$\begin{aligned} \boldsymbol{y} &= \mathbf{X}\boldsymbol{b} + \boldsymbol{\varepsilon}, \\ \mathbb{E}[\boldsymbol{\varepsilon}|\mathbf{X}] &= \mathbf{0}, \\ \mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top|\mathbf{X}] &= \sigma^2\boldsymbol{\Upsilon} = \boldsymbol{\Sigma}. \end{aligned} \tag{25}$$

Here, $\boldsymbol{y}$ is a $T \times 1$ vector, $\mathbf{X}$ is a $T \times (K + 1)$ matrix of explanatory variables, $\boldsymbol{b}$ is a $(K + 1) \times 1$ vector of regression coefficients, and $\boldsymbol{\varepsilon}$ is a $T \times 1$ vector of random errors. The $T \times T$ matrix $\boldsymbol{\Sigma}$ represents the covariance matrix of $\boldsymbol{\varepsilon}$. The primary difference between this model and the classical linear regression model is the introduction of the positive-definite matrix $\boldsymbol{\Upsilon}$. In the classical model, it is assumed that the errors $\boldsymbol{\varepsilon}$ have a constant variance and are independent, meaning that $\boldsymbol{\Upsilon}$ is simply the identity matrix $\mathbf{I}$.

However, for now, both assumptions of independence and homoscedasticity of $\boldsymbol{\varepsilon}$ can be relaxed, leading to two common issues, i.e., heteroscedasticity and autocorrelation. The introduction of $\boldsymbol{\Upsilon}$ is to reflect these properties of $\boldsymbol{\varepsilon}$. When only heteroscedasticity occurs, the covariance matrix $\sigma^2\boldsymbol{\Upsilon}$

takes the following form:

$$\sigma^2\boldsymbol{\Upsilon} = \sigma^2\begin{bmatrix} \upsilon_{11} & 0 & \cdots & 0 \\ 0 & \upsilon_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \upsilon_{TT} \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_T^2 \end{bmatrix}. \tag{26}$$

This matrix reflects the fact that the variance of $\boldsymbol{\varepsilon}$ can vary across observations (i.e., heteroscedasticity), but there is no correlation between the errors at different time points.

When there is only autocorrelation but no heteroscedasticity, the matrix $\sigma^2\boldsymbol{\Upsilon}$ takes the following form:

$$\sigma^2\boldsymbol{\Upsilon} = \sigma^2\begin{bmatrix} 1 & \rho_1 & \cdots & \rho_{T-1} \\ \rho_1 & 1 & \cdots & \rho_{T-2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{T-1} & \rho_{T-2} & \cdots & 1 \end{bmatrix}. \tag{27}$$

These two cases provide two typical examples of the matrix $\boldsymbol{\Upsilon}$. In more general cases, when both autocorrelation and heteroscedasticity are present simultaneously, the element in the i-th row and j-th column of the matrix $\boldsymbol{\Upsilon}$ is denoted by υ_{ij}.

If the matrix $\boldsymbol{\Upsilon}$ is known, the usual approach is to apply Generalized Least Squares (GLS) to estimate the regression coefficients $\boldsymbol{b}$. However, when $\boldsymbol{\Upsilon}$ is unknown, OLS is often the preferred method. The OLS estimtor is:

$$\hat{\boldsymbol{b}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{y} = \boldsymbol{b} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}. \tag{28}$$

However, heteroscedasticity or autocorrelation may affect the accuracy of standard errors.

The covariance of the OLS estimator $\hat{\boldsymbol{b}}$, denoted by $\mathbf{V}_{\text{OLS}}$, is:

$$\begin{aligned} \mathbf{V}_{\text{OLS}} &= \mathbb{E}\left[(\hat{\boldsymbol{b}} - \boldsymbol{b})(\hat{\boldsymbol{b}} - \boldsymbol{b})^\top|\mathbf{X}\right] \\ &= \mathbb{E}\left[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}|\mathbf{X}\right] \\ &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\sigma^2\boldsymbol{\Upsilon})\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} \\ &= \frac{1}{T}\left(\frac{1}{T}\mathbf{X}^\top\mathbf{X}\right)^{-1}\left(\frac{1}{T}\mathbf{X}^\top[\sigma^2\boldsymbol{\Upsilon}]\mathbf{X}\right)\left(\frac{1}{T}\mathbf{X}^\top\mathbf{X}\right)^{-1}. \end{aligned} \tag{29}$$

When there is no heteroskedasticity or autocorrelation in $\boldsymbol{\varepsilon}$, we have $\boldsymbol{\Upsilon} = \mathbf{I}$. Substituting this into (29) and performing some straightforward algebra gives the classic OLS expression for $\mathbf{V}_{\text{OLS}}$:

$$\mathbf{V}_{\text{OLS}} = \sigma^2\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}. \tag{30}$$

In practice, the sample residual variance s^2 is used as a substitute for the population variance σ^2, yielding an estimate of $\mathbf{V}_{\text{OLS}}$, denoted as $\hat{\mathbf{V}}_{\text{OLS}}$. The square roots of the diagonal elements of $\hat{\mathbf{V}}_{\text{OLS}}$

provide the standard errors for each of the regression coefficients, which are used for conducting t-statistics. However, if heteroskedasticity or autocorrelation exists in $\boldsymbol{\varepsilon}$, $\hat{\mathbf{V}}_{\text{OLS}}$ is no longer reliable.

In the expression of $\mathbf{V}_{\text{OLS}}$, aside from the factor $1/T$, the remaining terms form a product of three matrices. The first and third matrices are functions of the explanatory variables $\mathbf{X}$, so the key to adjusting for autocorrelation and heteroskedasticity lies in correctly estimating the middle matrix. For convenience, let $\mathbf{Q}$ represent this middle matrix:

$$\mathbf{Q} \equiv \frac{1}{T}\mathbf{X}^\top[\sigma^2\boldsymbol{\Upsilon}]\mathbf{X} = \frac{1}{T}\sum_{i=1}^{T}\sum_{j=1}^{T}\sigma_{ij}\boldsymbol{x}_i\boldsymbol{x}_j^\top. \tag{31}$$

Here, $\boldsymbol{x}_i = [x_{i1}, x_{i2}, \cdots, x_{iK+1}]^\top$, which is the transpose of the i-th row of $\mathbf{X}$ (note that it is not the i-th column of $\mathbf{X}$). Once we find an estimator of $\mathbf{Q}$ that accounts for autocorrelation and/or heteroskedasticity, we can use it to estimate the covariance matrix $\hat{\mathbf{V}}_{\text{OLS}}$ of $\hat{\boldsymbol{b}}$. Two well-known estimators for this purpose are the White (1980) estimator (for heteroskedasticity) and the Newey and West (1987) estimator (which adjusts for both heteroskedasticity and autocorrelation).

5.3 White and Newey-West Estimators

To estimate $\mathbf{Q}$, we utilize $\mathbf{X}$ and the sample residual vector from the regression, $\hat{\boldsymbol{\varepsilon}} = [\hat{\varepsilon}_1, \hat{\varepsilon}_2, \cdots, \hat{\varepsilon}_T]^\top$. When $\boldsymbol{\varepsilon}$ exhibits heteroscedasticity but no autocorrelation, $\mathbf{Q}$ simplifies to the following:

$$\mathbf{Q} = \frac{1}{T}\sum_{i=1}^{T}\sigma_i^2\boldsymbol{x}_i\boldsymbol{x}_i^\top. \tag{32}$$

White (1980) demonstrated that using $\mathbf{X}$ and $\hat{\boldsymbol{\varepsilon}}$, we can derive an asymptotically consistent estimator of $\mathbf{Q}$ (denoted as $\boldsymbol{S}_0$):

$$\mathbf{S}_0 = \frac{1}{T}\sum_{i=1}^{T}\hat{\varepsilon}_i^2\boldsymbol{x}_i\boldsymbol{x}_i^\top. \tag{33}$$

Using $\mathbf{S}_0$, we obtain the estimator for the covariance matrix of $\hat{\boldsymbol{b}}$:

$$\hat{\mathbf{V}}_{\text{OLS}} = \frac{1}{T}\left(\frac{1}{T}\mathbf{X}^\top\mathbf{X}\right)^{-1}\left(\frac{1}{T}\sum_{i=1}^{T}\hat{\varepsilon}_i^2\boldsymbol{x}_i\boldsymbol{x}_i^\top\right)\left(\frac{1}{T}\mathbf{X}^\top\mathbf{X}\right)^{-1}. \tag{34}$$

It can be simplified to:

$$\hat{\mathbf{V}}_{\text{OLS}} = T\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{S}_0\left(\boldsymbol{X}^\top\mathbf{X}\right)^{-1}. \tag{35}$$

This provides the adjusted covariance matrix of the OLS estimator under heteroscedasticity. It is known as the White heteroscedasticity-consistent estimator. Its advantage is that, even when the heteroscedasticity's structure or values are unknown, one can still make valid inferences based on the OLS results. This property is particularly important in practical applications where the nature of heteroscedasticity is unknown.

In anomaly testing, aside from heteroscedasticity, one must often account for the autocorrelation in $\boldsymbol{\varepsilon}$. A natural extension is:

$$\mathbf{S} = \frac{1}{T}\sum_{i=1}^{T}\sum_{j=1}^{T}\hat{\varepsilon}_i\hat{\varepsilon}_j\mathbf{x}_i\mathbf{x}_j^\top. \tag{36}$$

Although this expression might seem reasonable, it is incorrect for two reasons: (1) It involves a summation of T^2 terms, while the scaling factor is only $1/T$, meaning that $\mathbf{S}$ may not converge. (2) Even if $\mathbf{S}$ does converge, it may not be positive definite, leading to a non-positive-definite covariance matrix for $\hat{\boldsymbol{b}}$, which is clearly problematic.

To address these issues, Newey and West (1987) proposed a consistent estimator for $\mathbf{Q}$ that accounts for both heteroscedasticity and autocorrelation:

$$\mathbf{S} = \frac{1}{T}\sum_{t=1}^{T}\hat{\varepsilon}_t^2\mathbf{x}_t\mathbf{x}_t^\top + \frac{1}{T}\sum_{j=1}^{J}\sum_{t=j+1}^{T}w_j\hat{\varepsilon}_t\hat{\varepsilon}_{t-j}(\mathbf{x}_t\mathbf{x}_{t-j}^\top + \mathbf{x}_{t-j}\mathbf{x}_t^\top), \tag{37}$$

where $w_j = 1 - \frac{j}{1+J}$. The first term corresponds to the heteroscedasticity-only case, while the second term adjusts for autocorrelation. Here, J is the maximum lag for considering autocorrelation (with an adaptive algorithm for selecting J given in Newey and West 1994), and w_j is a weight that **diminishes** as lag j increases. Substituting this $\mathbf{S}$ into $\mathbf{V}_{\text{OLS}}$ provides the Newey-West heteroscedasticity and autocorrelation-consistent estimator of the covariance matrix:

$$\hat{\mathbf{V}}_{\text{OLS}} = T(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{S}(\mathbf{X}^\top\mathbf{X})^{-1}. \tag{38}$$

Taking the square root of the diagonal elements of the covariance matrix yields the standard errors of the regression coefficients. This estimator is widely used in empirical asset pricing studies. Nearly every paper analyzing factors or anomalies, when testing the significance of returns, will report Newey-West adjusted t-statistics or Newey-West adjusted standard errors. These terms mean that the standard errors and t-statistics are calculated using $\hat{\mathbf{V}}_{\text{OLS}} = T(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{S}(\mathbf{X}^\top\mathbf{X})^{-1}$.

Returning to the problem of anomaly testing in this section, the following steps outline how to apply the Newey-West adjustment to the standard errors of the regression coefficients in an OLS regression:

1. Use the anomaly returns as the dependent variable, and the returns of the K factors in the multi-factor model, along with an intercept (a total of $K+1$ variables), as the independent variables. Perform a time-series regression via OLS to obtain the residuals $\hat{\boldsymbol{\varepsilon}}$.

2. Using $\mathbf{X}$ and $\hat{\boldsymbol{\varepsilon}}$, compute the Newey-West adjusted $\hat{\mathbf{V}}_{\text{OLS}}$. The maximum lag J is determined using the formula from Newey and West (1994):

$$J = \left\lfloor 4 \times \left(\frac{T}{100}\right)^{2/9} \right\rfloor, \tag{39}$$

where $\lfloor \cdot \rfloor$ represents the floor function, which rounds down to the nearest integer.

3. Take the square root of the diagonal elements of $\hat{\mathbf{V}}_{\text{OLS}}$. These square roots are the standard errors for the regression coefficients $\hat{\boldsymbol{b}}$.

4. Locate the standard error for the intercept, which corresponds to the Newey-West adjusted standard error s.e.($\hat{\alpha}$) for $\hat{\alpha}$. Use $\hat{\alpha}$ and s.e.($\hat{\alpha}$) to calculate the t-statistic.

5.4 Cross-sectional Regression to Test Anomalies

Fama-MacBeth regression is also commonly employed to test anomalies. It allows us to test the (incremental) predictive power of the anomaly while controlling for other factors. During the regression, the anomaly variable, along with commonly used factor variables, is used as the explanatory variable, and the excess returns of individual assets are used as the dependent variable. Cross-sectional regression is conducted at each time t to estimate the return of the anomaly at time t, i.e., $\hat{\lambda}_t^\alpha$. Once we have the time series, we can calculate the mean and standard error of the anomaly returns (denoted as $\hat{\lambda}^\alpha$ and s.e.($\hat{\lambda}^\alpha$), respectively), and then perform a t-test. If the average return of the anomaly remains significant after controlling for factor variables, it is considered to earn α (i.e., abnormal return) that cannot be explained by those common factors.

Since the anomaly returns may exhibit heteroscedasticity and autocorrelation over time, the Newey-West adjustment can also be applied to obtain an accurate estimate of s.e.($\hat{\lambda}^\alpha$). However, how can we apply the Newey-West adjustment to a single return series? The correct method is to regress $\hat{\lambda}_t^\alpha$ on the constant 1 in a time-series regression (this actually demeans the return). The regression coefficient in this OLS is essentially the time-series mean $\hat{\lambda}^\alpha$, and the residuals are the deviations of the returns from the mean, reflecting the autocorrelation and heteroscedasticity in the anomaly returns. We can derive the Newey-West adjusted middle matrix. For a single time series, the middle matrix is a scalar (denoted as Q), and its estimator S is also a scalar:

$$S = \frac{1}{T}\left\{\sum_{t=1}^{T}\hat{\varepsilon}_t^2 + 2\sum_{j=1}^{J}\sum_{t=j+1}^{T} w_j \hat{\varepsilon}_t \hat{\varepsilon}_{t-j}\right\}, \tag{40}$$

where

$$\hat{\varepsilon}_t = \hat{\lambda}_t^\alpha - \mathbb{E}_T[\hat{\lambda}_t^\alpha], \quad w_j = 1 - \frac{j}{1+J}. \tag{41}$$

Since $X_t = 1$, we have $\mathbf{X}^\top\mathbf{X} = T$. Substituting S and $\mathbf{X}^\top\mathbf{X} = T$ into the Newey-West estimator, we obtain the variance for $\hat{\lambda}^\alpha$:

$$\hat{\sigma}^2_{\hat{\lambda}^\alpha} = \frac{S}{T}. \tag{42}$$

Taking the square root yields the Newey-West adjusted standard error for $\hat{\lambda}^\alpha$: s.e.($\hat{\lambda}^\alpha$) $= \sqrt{S/T}$. Finally, we use $\hat{\lambda}^\alpha$ and s.e.($\hat{\lambda}^\alpha$) to calculate the t-statistic and test the significance of anomaly's average returns.

It is important to note that the Newey-West adjustment described above applies to all independent variables in the regression, not just to the anomaly variable. If the independent variable is the anomaly variable, the test checks whether the expected anomaly returns are significant. If the independent variable is a factor variable, it tests the significance of the expected factor returns. Additionally, the Newey-West adjustment can also be applied to the sorting method. This means that after obtaining the factor return time series $\{\hat{\lambda}_{kt}\}$ through portfolio sort, we also can calculate Newey-West adjusted standard error.

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